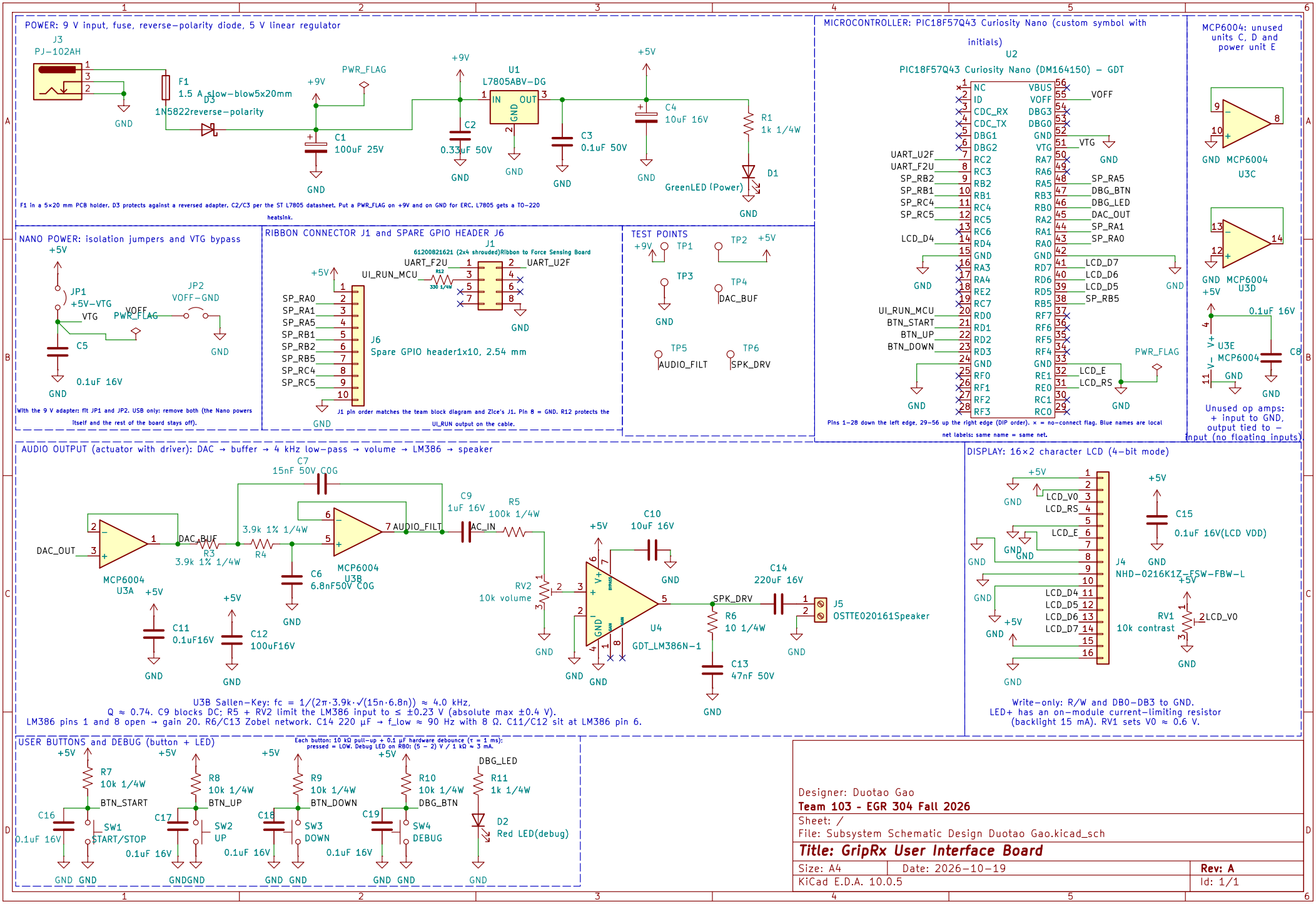

POWER: 9 V input, fuse, reverse-polarity diode, 5 V linear regulator
J3
PJ-102AH
F1
1.5 A slow-blow5x20mm
D3
1N5822reverse-polarity
PWR_FLAG
U1
L7805ABV-DG
C1
100uF 25V
C2
0.33uF 50V
C3
0.1uF 50V
C4
10uF 16V
R1
1k 1/4W
D1
GreenLED (Power)
GND
+9V
+5V
F1 in a 5×20 mm PCB holder. D3 protects against a reversed adapter. C2/C3 per the ST L7805 datasheet. Put a PWR_FLAG on +9V and on GND for ERC. L7805 gets a TO-220 heatsink.
MICROCONTROLLER: PIC18F57Q43 Curiosity Nano (custom symbol with initials)
U2
PIC18F57Q43 Curiosity Nano (DM164150) – GDT
UART_U2F
UART_F2U
SP_RB2
SP_RB1
SP_RC4
SP_RC5
LCD_D4
UI_RUN_MCU
BTN_START
BTN_UP
BTN_DOWN
VOFF
VTG
SP_RA5
DBG_BTN
DBG_LED
DAC_OUT
SP_RA1
SP_RA0
LCD_D7
LCD_D6
LCD_D5
SP_RB5
LCD_E
LCD_RS
PWR_FLAG
Pins 1–28 down the left edge, 29–56 up the right edge (DIP order). × = no-connect flag. Blue names are local net labels; same name = same net.
MCP6004: unused units C, D and power unit E
GND MCP6004
U3C
GND MCP6004
U3D
GND MCP6004
U3E
0.1uF 16V
C8
Unused op amps: + input to GND, output tied to − input (no floating inputs).
NANO POWER: isolation jumpers and VTG bypass
JP1
+5V–VTG
JP2
VOFF–GND
C5
0.1uF 16V
With the 9 V adapter: fit JP1 and JP2. USB only: remove both (the Nano powers itself and the rest of the board stays off).
RIBBON CONNECTOR J1 and SPARE GPIO HEADER J6
J1
61200821621 (2x4 shrouded)Ribbon to Force Sensing Board
R12
330 1/4W
J6
Spare GPIO header1x10, 2.54 mm
SP_RA0
SP_RA1
SP_RA5
SP_RB1
SP_RB2
SP_RB5
SP_RC4
SP_RC5
J1 pin order matches the team block diagram and Zice's J1. Pin 8 = GND. R12 protects the UI_RUN output on the cable.
TEST POINTS
TP1
TP2
TP3
TP4
DAC_BUF
TP5
AUDIO_FILT
TP6
SPK_DRV
AUDIO OUTPUT (actuator with driver): DAC → buffer → 4 kHz low-pass → volume → LM386 → speaker
C7
15nF 50V C0G
C9
1uF 16V
R5
100k 1/4W
C10
10uF 16V
DAC_OUT
DAC_BUF
R3
3.9k 1% 1/4W
R4
3.9k 1% 1/4W
C6
6.8nF50V C0G
MCP6004
U3A
MCP6004
U3B
AUDIO_FILT
AC_IN
RV2
10k volume
U4
GDT_LM386N-1
C11
0.1uF16V
C12
100uF16V
C13
47nF 50V
C14
220uF 16V
SPK_DRV
R6
10 1/4W
J5
OSTTE020161Speaker
U3B Sallen-Key: fc = 1/(2π·3.9k·√(15n·6.8n)) ≈ 4.0 kHz, Q ≈ 0.74. C9 blocks DC; R5 + RV2 limit the LM386 input to ≤ ±0.23 V (absolute max ±0.4 V). LM386 pins 1 and 8 open → gain 20. R6/C13 Zobel network. C14 220 µF → f_low ≈ 90 Hz with 8 Ω. C11/C12 sit at LM386 pin 6.
DISPLAY: 16×2 character LCD (4-bit mode)
J4
NHD-0216K1Z-FSW-FBW-L
C15
0.1uF 16V(LCD VDD)
RV1
10k contrast
LCD_V0
LCD_RS
LCD_E
LCD_D4
LCD_D5
LCD_D6
LCD_D7
Write-only: R/W and DB0–DB3 to GND. LED+ has an on-module current-limiting resistor (backlight 15 mA). RV1 sets V0 ≈ 0.6 V.
USER BUTTONS and DEBUG (button + LED)
Each button: 10 kΩ pull-up + 0.1 µF hardware debounce (τ = 1 ms): pressed = LOW. Debug LED on RB0: (5 − 2) V / 1 kΩ ≈ 3 mA.
R7
10k 1/4W
BTN_START
SW1
START/STOP
C16
0.1uF 16V
R8
10k 1/4W
BTN_UP
SW2
UP
C17
0.1uF 16V
R9
10k 1/4W
BTN_DOWN
SW3
DOWN
C18
0.1uF 16V
R10
10k 1/4W
DBG_BTN
SW4
DEBUG
C19
0.1uF 16V
R11
1k 1/4W
DBG_LED
D2
Red LED(debug)
Designer: Duotao Gao
Team 103 - EGR 304 Fall 2026
Sheet: /
File: Subsystem Schematic Design Duotao Gao.kicad_sch
Title: GripRx User Interface Board
Size: A4
Date: 2026-10-19
Rev: A
KiCad E.D.A. 10.0.5
Id: 1/1